

# CodeBugFinder - Complete Project Documentation

## Project Overview

CodeBugFinder is a Python-based Static Code Analysis Tool developed using Abstract Syntax Tree (AST). The tool scans Python source code and detects security vulnerabilities, coding issues, and code quality problems.

## Objectives

- Detect bugs automatically.
- Improve code quality.
- Generate professional reports.
- Provide a dashboard for visualization.

## Technology Stack

Python, AST (Abstract Syntax Tree), Streamlit, JSON, CSV, File Handling.

## Project Structure

```
CodeBugFinder/  
├── scanner/  
├── reports/  
├── output/  
├── app.py  
├── main.py  
└── README.md
```

## Implemented Rules

CBF001 - Hardcoded Password  
CBF002 - Possible Infinite Loop  
CBF003 - Debug Print Statement  
CBF004 - Empty Exception Handler  
CBF005 - TODO Comment Found  
CBF006 - Hardcoded API Key  
CBF007 - Bare Exception Detected  
CBF008 - Wildcard Import Detected

## Reports Generated

- TXT Report
- CSV Report
- JSON Report
- Streamlit Dashboard

## Dashboard Features

- Total Issue Count
- Severity Metrics (HIGH/MEDIUM/LOW)
- Issue Table
- Filtering and Visualization Ready

## How It Works

1. Read Python files.
2. Parse source code into AST.
3. Apply detection rules.
4. Collect findings.
5. Generate reports.
6. Display dashboard.

## Resume Description

Developed CodeBugFinder, a Python-based static code analysis tool using AST. Implemented custom security and code-quality detection rules, folder-level scanning, report generation (TXT, CSV, JSON), and a Streamlit dashboard for visualization.

## Future Enhancements

- HTML Reports
- Duplicate Code Detection
- Multi-language Support
- Cloud Deployment
- AI-based Fix Suggestions